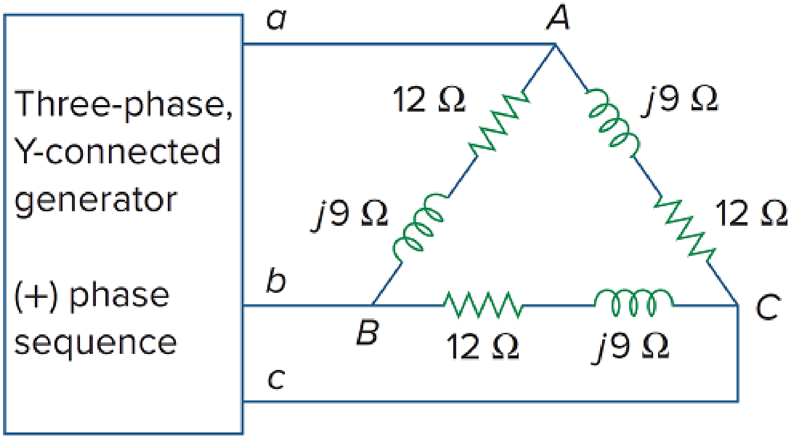
Problem

If $\mathbf{V}_{an} = 220\angle 60^\circ\ \mathrm{V}$ in the network of Fig. 12.49, find the load phase currents **I***AB*, **I***BC*, and **I***CA*.

Figure 12.49



Step-by-step solution

Step 1 of 5

Draw the following circuit diagram:

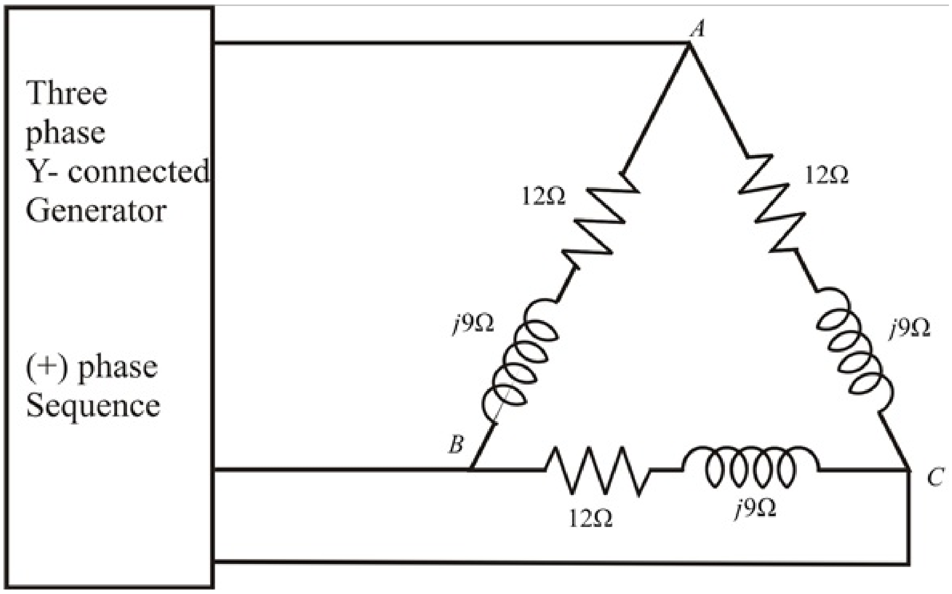


Figure 1

Step 2 of 5

Consider the voltage $\mathbf{V}_{an} = 220\angle 60° \text{ V}$.

Write the expression for the line voltage $\mathbf{V}_{AB}$.

$$\begin{aligned}\mathbf{V}_{AB} &= \sqrt{3}\mathbf{V}_{an}\angle 30° \\ &= \sqrt{3}\times 220\angle(60°+30°) \\ &= 381.05\angle 90° \text{ V}\end{aligned}$$

Therefore, the value of line voltage $\mathbf{V}_{AB}$ is $381.05\angle 90° \text{ V}$.

Step 3 of 5

Calculate the value of the phase current $\mathbf{I}_{AB}$.

Write the expression for phase currents $\mathbf{I}_{AB}$.

$$\begin{aligned}\mathbf{I}_{AB} &= \frac{\mathbf{V}_{AB}}{\mathbf{Z}_{AB}} \\ &= \frac{381.05\angle 90°}{12+j9} \\ &= \frac{381.05\angle 90°}{15\angle 36.86°} \\ &= 25.40\angle 53.13° \text{ A}\end{aligned}$$

Therefore, the value of line currents $\mathbf{I}_{AB}$ is $\boxed{25.40\angle 53.13° \text{ A}}$.

Step 4 of 5

Calculate the value of the phase current $\mathbf{I}_{BC}$.

Write the expression for phase currents $\mathbf{I}_{BC}$ for a balanced system.

$$\begin{aligned}\mathbf{I}_{BC} &= \mathbf{I}_{AB}\angle -120° \\ &= 25.40\angle(53.13° - 120°) \\ &= 25.40\angle -66.87° \text{A}\end{aligned}$$

Therefore, the value of phase current $\mathbf{I}_{BC}$ is $\boxed{25.40\angle -66.87° \text{A}}$.

Step 5 of 5

Calculate the value of the phase current $\mathbf{I}_{CA}$.

Write the expression for phase currents $\mathbf{I}_{CA}$ for a balanced system.

$$\begin{aligned}\mathbf{I}_{CA} &= \mathbf{I}_{AB}\angle 120° \\ &= 25.40\angle(53.13° + 120°) \\ &= 25.40\angle 173.13° \text{A}\end{aligned}$$

Therefore, the value of phase current $\mathbf{I}_{CA}$ is $\boxed{25.40\angle 173.13° \text{A}}$.